

Analyzing Diagnostic Breast Cancer Data

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Introduction

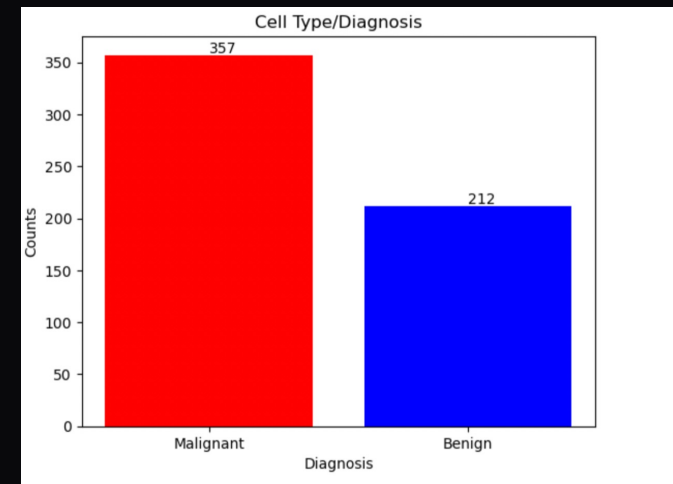
- The data I chose: Breast Cancer Wisconsin (Diagnostic) from the UCI Machine Learning Repository
- How is the data collected? (FNAs)
- Features include: radius, perimeter, area, texture, and more.
- 569 patients, 30 features
- Variables of interest: mean texture, worst area, worst smoothness

Research Goals:

1. Is there a relationship between certain aspects of a tumor and its ultimate diagnosis? (featured variables)
2. Are malignant (cancerous) tumors significantly different from benign tumors in size?
3. How can this data be used to identify commonalities or unique features of a malignant tumor?

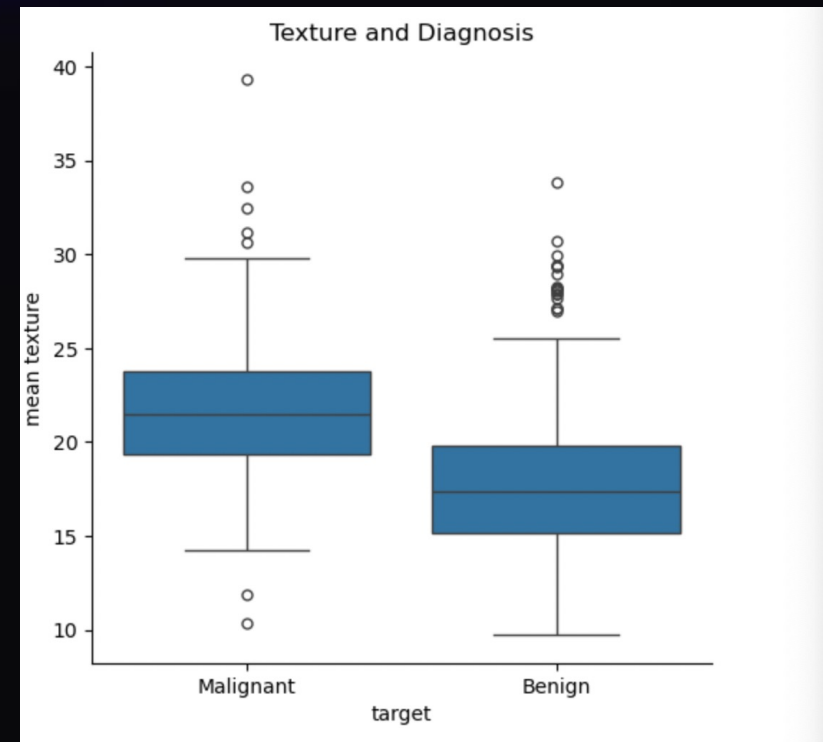
Cleaning/Preprocessing

1. The data contained no missing values
2. The diagnostic variable 'target' had to be converted from numerical to categorical
3. The data contained 357 malignant cases and 212 benign cases



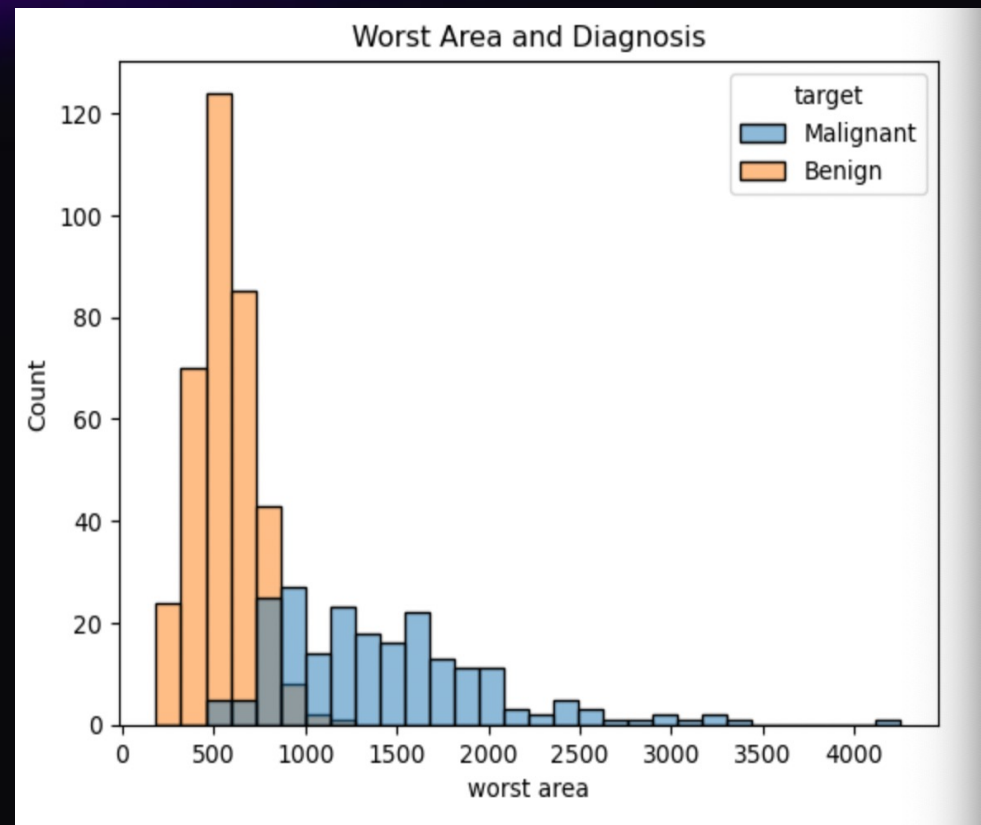
Texture

- Texture: standard deviation of gray-scale values
- We can observe that the median for malignant tumors is higher than the median of benign.
- This suggests an association between tumor texture and cancer.



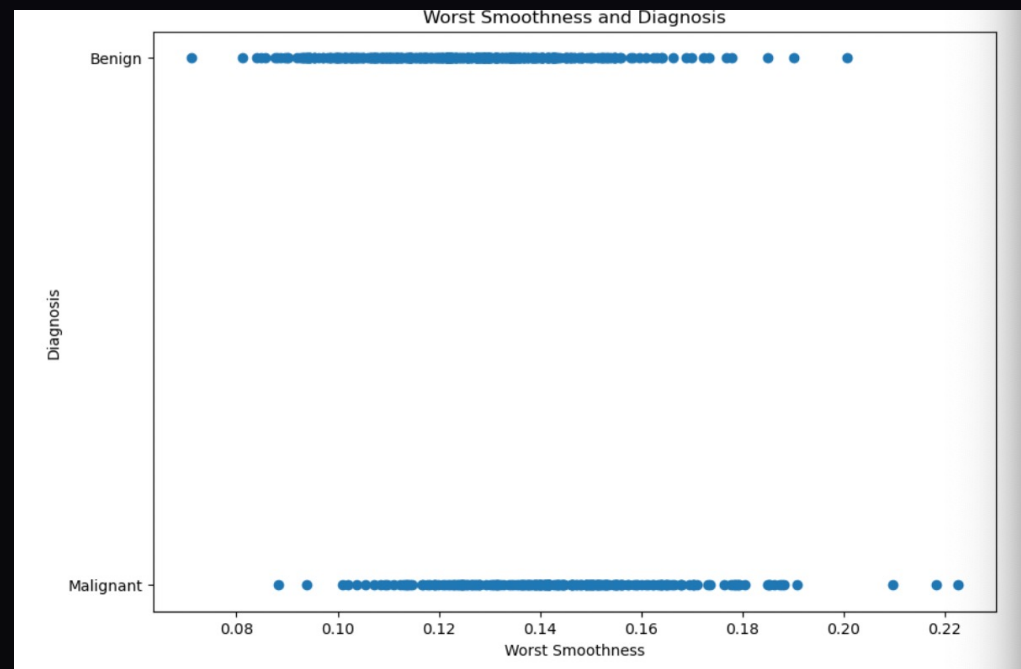
Area

- The area for malignant tumors is more often larger while most benign tumors have a smaller area.
- Larger tumor size may indicate a higher likelihood of malignancy



Smoothness

- Smoothness refers to the variation in radius lengths (how uneven it is)
- Looking at cluster patterns, malignant tumors appear to have greater variations in radius. (more jagged edges, less even)



Research Goals Revisited

1. Malignant tumor FNAs tend to be larger, more textured and less even in comparison with benign tumor FNAs
2. Size is a contributing factor to evaluating a tumor for diagnosis.
3. Unique features that stand out when determining a diagnosis are area and irregularities in texture and smoothness.

Business and societal relevance:

- can help improve diagnosis accuracy/ increase detection speed
- efficient diagnosis can reduce costs to companies, hospitals, and patients

Mindfulness

- Real people, not just numbers
- Taking time to be efficient

Next Steps

- Creating a detecting algorithm/test
- Adding more personal features